

Entry #		Date	Phase	Human Initiator	Category	Context [1]	Prompt/Excerpt [2]	AI Summary [3]	Human Evaluation [4]	Final Decision [5]	Testing Verification [6]	Rejected AI? (Y/N) [7]	Professor/TA Comments
1		3/4/26	1	Anthony Frialde	Architecture	I wanted to know what the difference is going to be between the battleship game that I made before and the new one we are making with multiplayer	I want to make a battleship game like the one I made before but it needs to support multiplayer with many instances of games going on at the same time. It needs to have a customizable grid size. It needs to be persistent like my old battleship game. It needs to support player stats as a leaderboard system. I do not want you to code anything yet I just want you to layout the structure of how we would do this.	pretty much just talks about how to setup a multiplayer environment to where you have a status and id for a multiplayer session and players must make a username before joining a match via a client, release() call, which would have exhausted the connection pool.	I think what it output was good I was not looking for any sort of guidance on how chatgpt thinks the game should be setup	We haven't implemented anything from this prompt I think it is just good to know personally and what we can look to in the future when we begin to start doing things	Again there is nothing to test, just looking for some guidance	Y	
2		3/4/26	1	Anthony Frialde	Feature	When first building the API I needed to create the initial routes for the Battleship game. I wanted to confirm what endpoints were typically needed for starting and managing a game.	What are some common REST endpoints for a simple battleship game API?	The AI suggested endpoints for creating games, joining games, placing ships, and making moves.	This helped confirm that my API structure made sense and covered the basic game actions.	I kept this in mind as I moved forward with the process of designing the project.	No testing needed just idea	N	
3		3/5/26	1	Christian Johnston	Bug Fix	The server was returning 500 errors during game creation on Render. We weren't sure if it was due to code logic or a mismatch with the database schema.	My create game route is returning a 500 error when I try to run my tests, where do you think this is coming from?	Identified a potential null constraint violation occurring on the current_turn_index column during insertion.	The AI was right about the violation, but it suggested an unnecessary schema wipe and rewrite	Rejected the schema rewrite, and instead modified the SQL INSERT statement to set a default value of 0	Verified by successfully creating games through the Render URL and observing 201 responses	Y	
4		3/8/26	1	Christian Johnston	Bug Fix	I was getting immediate database connection timeouts when trying to deploy to Render. I wasn't certain about the requirements for external PostgreSQL connections to Render	What about my Postgre/SQL configuration is not working with Render's environment setup	Recommended adding rejectUnauthorized: false to the SSL configuration in the pg Pool constructor.	This was the right fix to handle the Render's authentication certificates	Implemented the SSL config in db.js while keeping the connection string as an environment variable	Verified by checking Render's service dashboard and seeing "Connected to Database"	N	
5		3/9/26	1	Christian Johnston	Architecture	Multi-step database inserts (creating a game then joining the creator) were failing during high-latency periods. The risk was creating games without players.	Can you help me use SQL transactions for creating a game	Recommended wrapping the queries in BEGIN, COMMIT, and ROLLBACK statements to ensure atomicity	The concept was correct, but the provided code didn't have a client, release() call, which would have exhausted the connection pool.	Rejected the AI's implementation and had it make a more robust transaction handler	Verified by forcing a failure on the second query and confirming the first query was successfully rolled back	Y	
6		3/9/26	1	Anthony Frialde	Bug Fix	I was getting immediate database connection timeouts when trying to deploy to Render. I suspected the issue was related to the database connection configuration.	Why would a Node.js API fail to connect to a PostgreSQL database?	The AI suggested checking the connection string and environment variables.	This helped identify that my rejectUnauthorized variable was not being loaded correctly.	I fixed the database connection configuration.	I ran a test query and confirmed the database connection worked.	N	
7		3/9/26	1	Christian Johnston	Testing	Local tests were failing because the database IDs were incrementing too high after multiple resets. I was expecting 1 but getting 20 for the IDs	My tests are failing because they expect player_id 1 but the database gives me 20	Recommended using TRUNCATE and RESTART IDENTITY instead of DELETE to reset the auto-increment counters	This is a much better solution for testing than manually tracking high IDs	Rejected the AI's TRUNCATE command and kept the DELETE command for safety, but modified the tests to dynamically use the ID returned from the API	Verified by updating the Python test script to capture and reuse player_id from the creation response.	Y	
8		3/9/26	1	Christian Johnston	Feature	We needed a way to determine who should move next in a multiplayer game. I wasn't sure I should use the join_order or the player's ID. A 404 error happened when fetching stats for a new player. I suspected the INSERT query was not returning the primary key correctly	How should I determine which player goes first and whose turn is next?	Recommended adding a turn_order column to the game_players table to track player sequence IDs	Using a relative turn order is more flexible than relying on the player sequence IDs	Implemented turn_order (0 or 1) and the joiner is order 1	Verified that the game creator is always 0 and the joiner is order 1	N	
9		3/10/26	1	Christian Johnston	Bug Fix	While connecting the API to the database I ran into an error when trying to query tables. I suspected the issue was related to the database connection configuration.	Why am I getting a 404 when I try getting stats for a new player?	Suggested checking if the RETURNING clause in the SQL query matched the column names in the schema exactly	The AI was correct. The code was expected id but the schema called directly for player_id	Updated all controller methods to use result.rows[0].player_id to keep names consistent with the database	Verified by creating a player and successfully fetching their stats in a single sequence.	N	
10		3/10/26	1	Anthony Frialde	Testing	While building the test endpoints I needed to verify that the API behaved consistently when resetting the game state. I wanted to confirm the best way to reset database tables for deterministic testing.	What is a good way to reset database tables in a test API so each test starts with a clean state?	The AI suggested deleting relevant tables before running tests.	This approach worked well for the testing endpoints and kept the database state predictable.	I implemented a reset endpoint that clears the relevant tables before tests run.	I ran multiple test cycles and verified the API returned the same results each time.	N	
11		3/10/26	1	Christian Johnston	Feature	We needed a test-only route to reveal ship locations without exposing secret game data to players in a normal game	How do I make a route that shows the full board only when in test mode?	Suggested creating a specific /api/test/ router and using middleware to verify the x-test-mode header	This effectively isolated the test functions from the standard game logic	Implemented a verifyTestMode middleware and mapped it to a new reveal Board controller.	Verified that the route returns ship data only when the correct secret key is provided in the headers	N	
12		3/10/26	1	Christian Johnston	Testing	While testing the createGame endpoint, I realized my validation wasn't strict enough. I needed to make sure that the server rejected invalid numbers for the grid size specifically to match the project requirements	How do I validate that the grid size is between 5 and 15 in my controller properly?	Suggested an if block that verifies the grid_size is within range, and returns a 400 error if it isn't	The logic was correct, but I had to add my own check to make sure the value was actually provided in the request at all	Added a validation check in game Controller.js that verifies all required fields exist and that the grid_size falls within the 5-15 range	Verified by sending a grid_size of 4 and confirming the server responded with a 400 error and an error message	N	
13		3/10/26	1	Anthony Frialde	Bug Fix	While working on the getStats endpoint I noticed the response was returning incorrect values when multiple games existed in the database. I needed to figure out how to properly calculate statistics across the tables.	How should I structure a SQL query to calculate game statistics from multiple tables in a battleship API?	The AI suggested using aggregate functions like COUNT and GROUP BY to calculate game statistics such as wins and games played.	The suggestion helped clarify how to structure the query. Some parts were unnecessary since my schema already limited certain values.	I updated the SQL query inside the getStats function to correctly aggregate statistics.	I created multiple games in the database and verified the stats endpoint returned correct totals.	N	
14		3/10/26	1	Christian Johnston	Bug Fix	Joining a game was failing on Render but working locally. I wasn't sure why	I'm getting a 500 error on join game only on Render, but not locally. Why is this happening?	Identified that the production database was likely missing the current_turn_index column due to an incomplete migration.	The AI was exactly right, the schema in our project had drifted from the local schema.sql	Manually executed the missing ALTER TABLE commands via the Render database console to get the environments in sync	Verified by successfully joining a game on the production server	N	
15		3/10/26	1	Anthony Frialde	Bug Fix	While running the API locally I ran into issues where routes were not being recognized because of naming and casing differences between files. This caused the testController and testRoutes imports to fail.	For some reason my testcontroller file is not working in render all I did was change the casing what should I do.	The AI suggested me to change the file to a different name completely temporarily and then change it back and then push the commit to render	The recommendation worked I think it was because of a caching issue.	I kept the name and the issue was fixed	I pushed the new commit to render and it worked.	N	
16		3/10/26	1	Anthony Frialde	Feature	When preparing the API for deployment I needed to connect the project to a PostgreSQL database hosted on Render. I wanted to confirm the correct way to configure the database connection in a Node.js project.	How do you connect a Node.js API to a PostgreSQL database hosted on Render?	The AI recommended using environment variables to store the database connection string and using the pg library to manage the connection.	This suggestion was correct and matched typical Node.js deployment practices. It also helped avoid hardcoding sensitive database credentials.	I configured the database connection using environment variables and updated the db.js configuration accordingly.	I ran API requests that inserted and retrieved data to confirm the Render PostgreSQL database was working correctly.	N	
17		3/10/26	1	Anthony Frialde	Feature	While building the game API I needed to create an endpoint that would start a new game and create the required database records. I wanted to make sure the function created the game properly.	What should happen in a createGame endpoint for a battleship API?	The AI suggested inserting a new game record and linking players to that game.	The suggestion matched the structure I was planning for the API.	I implemented the createGame function so it creates a game entry in the database.	I tested the endpoint and confirmed that a new game record was created	N	
18		3/10/26	1	Christian Johnston	Bug Fix	Initial database queries were failing because the application couldn't find the DATABASE_URL	Why is my db.js not picking up the environment variables?	The AI suggested checking the dotenv package and calling config() at the very top of the entry file	This was necessary for local development and I overlooked this in the initial setup	Added require('dotenv').config(); to the top of db.js	Verified by successfully connecting to the local PostgreSQL instance using .env variables	N	
19		3/10/26	1	Anthony Frialde	Optimization	During development I noticed some endpoints were making multiple database calls when placing ships or retrieving game data. I wanted to check whether combining queries could improve performance.	Is it better to run multiple small SQL queries or combine them into one query in a REST API?	The AI suggested combining queries when possible to reduce database round trips.	The suggestion was correct but performance was not a major concern for phase 1	I kept the existing query structure since it was simpler and easier to debug.	The endpoints were retested to confirm they still returned correct results.	Y	
20		3/10/26	1	Christian Johnston	Bug Fix	The /api/reset route was failing with foreign key constraint errors	How do I clear all data for a reset without getting these foreign key errors?	Provided the correct deletion sequence to remove dependent rows before parent rows	The logic was correct and followed the relational integrity rules	Implemented the specific deletion order in the app.post("/api/reset") route	Verified by running the reset endpoint and confirming all tables were cleared without errors	N	
21		3/11/26	1	Anthony Frialde	Testing	While testing the API I wanted to make sure the server restarted correctly after changes. I needed a simple way to automatically restart the server during development.	How can I automatically restart a Node.js server during development?	The AI suggested using tools like nodemon.	This was helpful and made development easier.	I used nodemon to restart the server automatically.	I confirmed the server restarted after code changes.	N	
22		3/11/26	1	Christian Johnston	Refactor	The fireShot logic was hardcoding "miss" as a result. We needed to prepare the codebase for actual hit detection in Phase 2	I'm setting up fireShot, should I do the hit detection now?	Suggested implementing a placeholder result now so the API contract is met, while keeping the SQL query ready for coordinates	This was good advice to keep phase 1 simple while making room for the future implementation of phase 2	Implemented the route to return miss by default but record the exact row and col in the moves table	Verified by checking that moves are correctly recorded in the database	N	
23		3/11/26	1	Christian Johnston	Testing	We needed a way to test the API locally without constantly deploying to Render	Help me turn these tests (from the autograder) into a python test script so I can test locally	Generated a python script using the requests library to hit the API endpoints in the correct sequence	The script functioned but lacked a way to easily switch between local and remote environments	Modified the script to use a BASE_URL variable, allowing for me to change my tests between localhost and Render	Verified by running the script against the local server and confirming it passed all the requirements	N	

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24		3/11/26	1	Anthony Frialde	Testing	While building some endpoints I realized I wasn't always sure what HTTP status codes I should return for different situations. I wanted to make sure the API responses followed common REST practices.	What HTTP status codes should a REST API return for successful requests and errors?	The AI suggested using 200 for successful requests, 400 for bad input, and 500 for server errors.	This was helpful because it clarified which status codes should be used in different cases.	I updated some endpoints to return clearer status codes for errors and successful responses.	I tested the endpoints and confirmed the correct status codes were returned.	N	
25		3/11/26	1	Anthony Frialde	Testing	While verifying the API endpoints I wanted a quick way to send requests without building a frontend interface. I needed a simple tool for manually testing the API.	What tools can I use to manually test a REST API during development?	The AI suggested using tools like Postman or curl to send HTTP requests.	This was helpful because it allowed me to quickly test endpoints while building the API without having to use the autograder.	I used Postman to test the API requests during development.	I verified the API responses by sending requests through Postman.	N	
26		3/11/26	1	Anthony Frialde	Bug Fix	While running the server locally I encountered an error when trying to start the API. I suspected the issue was related to how modules were being exported from one of the controller files.	Why would a Node.js controller function not be recognized when imported into a route file?	The AI suggested checking that the function was exported correctly and that the import path was accurate.	This helped me quickly locate the issue since the function name in the export did not match the import.	I corrected the export statement in the controller file.	After restarting the server, the route worked correctly.	N	
27		3/12/26	1	Anthony Frialde	Testing	While testing my API I wanted to make sure errors were handled properly when invalid input was sent. I needed a quick way to simulate bad requests.	How can I test invalid requests in a REST API?	The AI suggested sending requests with missing fields or incorrect values.	This helped verify that the API returned the correct error responses.	I tested the endpoints with invalid inputs to confirm error handling.	The API returned the expected error codes.	N	
28		3/12/26	1	Anthony Frialde	Refactor	The controller file started getting messy because it contained both API logic and database queries. I wanted to make the code easier to read and maintain.	How should I organize a Node.js API so database queries are separated from route logic?	The AI suggested separating the project into controllers, services, and database modules.	The recommendation was good but a full service layer seemed unnecessary for this small project.	I moved database queries into a separate db.js file but kept most logic in the controllers.	I retested the endpoints to confirm the refactor didn't break ship placement.	N	
29		3/15/26	1	Christian Johnston	Bug Fix	I was getting a 500 error when trying to use the test-mode board reveal. I realized it was because I was passing the player ID as a body parameter instead of a URL parameter which didn't match the route I had defined.	GET /api/test/games/{id}/board(player_id) is not working properly.	Pointed out to me that the route definition in test Routes.js needed to include :player_id as a variable in the URL path to match our contract.	This was a simple routing error where I was trying to read from req.body on a GET request.	Updated the route to /id/board/:player_id and changed the controller to pull the ID from req.params	Verified by calling the URL with a specific player ID and successfully receiving the ship coordinates in the response.	N	
30		3/15/26	1	Christian Johnston	Optimization	The getGame response was returning the creator's ID, but I also needed it to show how many players were currently in the game so the frontend would know if it was full.	I need the game state to show how many players have joined	Recommended using a subquery in the SELECT statement for the games table to count the number of rows in game_players for that specific game	This was more efficient than fetching the game and then making another database call just to count players	Updated the get Game logic in game Controller.js to include an active_players count using an SQL subquery	Verified by checking the JSON response for a game and confirming active_players changed from 1 to 2 after a second player joined.	N	

[1] Describe the engineering situation that led to the AI interaction.
Include constraints and uncertainty.
2–5 sentences. Do not summarize AI output.

[2] Include only the relevant portion of your prompt.
Do not paste entire conversations.

[3] Summarize the AI recommendation in your own words.
Do not paste the full response.

[4] Critically evaluate the AI output.
What was correct? What was wrong? What was rejected?

[5] What did we actually implement — and why?

[6] Explain how correctness was verified.
Include regression testing if applicable.

[7] At least one “Yes” per phase is expected.